Healthcare Management System using MySQL

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Display departments having more than 5 appointments.

```sql
select department,
count(*) AS TotalAppointments
FROM appointments
group by department
having count(*) > 5;
```

department	TotalAppointments
Cardiology	8
Dermatology	8
Orthopedics	6
General	8
ENT	6

- **Identifies departments with high appointment volume.**
- **Uses aggregate results to compare departments.**
- **Provides a summary of appointment distribution by department.**

Display the total consultation fee for each department.

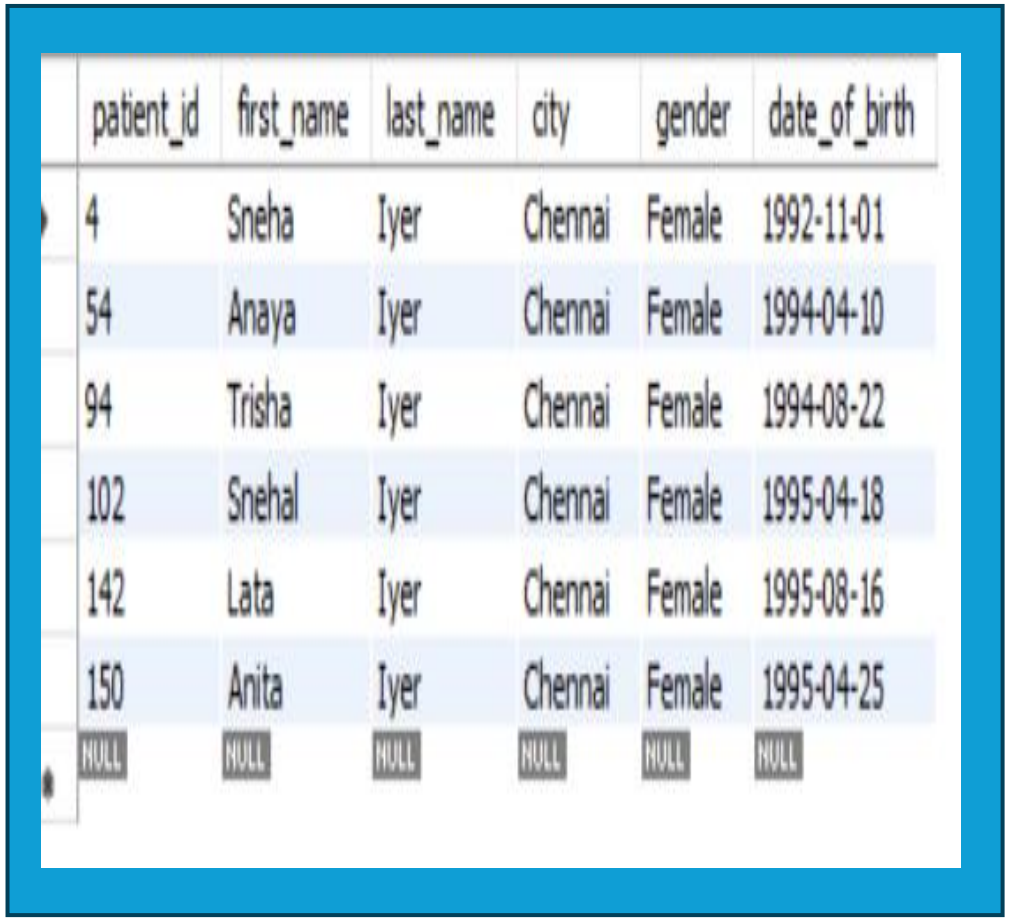
```sql
select appointment_id,department,consultation_fee,
sum(consultation_fee) over(partition by department) AS total_fee
FROM appointments;
```

appointment_id	department	consultation_fee	total_fee
101	Cardiology	800	6750
109	Cardiology	800	6750
134	Cardiology	850	6750
127	Cardiology	900	6750
115	Cardiology	850	6750
121	Cardiology	800	6750
140	Cardiology	900	6750
146	Cardiology	850	6750
102	Dermatology	500	4250
137	Dermatology	500	4250
111	Dermatology	550	4250
131	Dermatology	550	4250
116	Dermatology	500	4250
122	Dermatology	600	4250
143	Dermatology	500	4250
149	Dermatology	550	4250
150	ENT	650	3750
107	ENT	600	3750
135	ENT	650	3750
129	ENT	600	3750
119	ENT	650	3750

- **This query finds the total consultation fee for each department.**
- **It shows every appointment with its department and consultation fee.**
- **The total fee is repeated for all appointments in the same department.**

Display female patients from chennai whose age is less than 35.

```
select *from patients where gender = 'Female' and city = 'chennai'
 and TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) < 35;
```



patient_id	first_name	last_name	city	gender	date_of_birth
4	Sneha	Iyer	Chennai	Female	1992-11-01
54	Anaya	Iyer	Chennai	Female	1994-04-10
94	Trisha	Iyer	Chennai	Female	1994-08-22
102	Snehal	Iyer	Chennai	Female	1995-04-18
142	Lata	Iyer	Chennai	Female	1995-08-16
150	Anita	Iyer	Chennai	Female	1995-04-25
NULL	NULL	NULL	NULL	NULL	NULL

- **Retrieves patient records from the patients table.**
- **Uses TIMESTAMPDIFF() to calculate the patient's age.**
- **Shows only patients who satisfy all the given conditions.**

Display doctors whose total consultation fee is above the overall average consultation fee.

```sql
select doctor_name,SUM(consultation_fee) AS total_fee
from appointments
group by doctor_name
having SUM(consultation_fee) >
(
    select avg(consultation_fee)
    FROM appointments
);
```

doctor_name	total_fee
Dr. Rao	1650
Dr. Iyer	1750
Dr. Verma	2400
Dr. Patel	800
Dr. Nair	1100
Dr. Joshi	1100
Dr. Pillai	950
Dr. Malhotra	850
Dr. Jain	750
Dr. Mishra	900
Dr. Chopra	800
Dr. Kapoor	1850
Dr. Saxena	1150
Dr. Shah	900
Dr. Srivastava	750
Dr. Kulkarni	950
Dr. Naik	850
Dr. Shetty	1300
Dr. Arora	1200

- The query finds the total consultation fee for each doctor.
- It compares each doctor's total fee with the average consultation fee.
- It shows only the doctors whose total fee is above the average.

Count the total number of appointments for each patient.

```sql
select  p.patient_id,p.first_name,
COUNT(a.appointment_id) AS total_appointments
from patients p left join appointments a
ON p.patient_id = a.patient_id
group by p.patient_id, p.first_name;
```

patient_id	first_name	total_appointments
1	Ravi	1
2	Priya	1
3	Arjun	1
4	Sneha	1
5	Amit	1
6	Neha	1
7	Rahul	1
8	Anjali	1
9	Kiran	1
10	Vikram	1
11	Pooja	1
12	Suresh	1
13	Deepak	1
14	Meena	1
15	Raj	1
16	Simran	1
17	Nikhil	1
18	Divya	1

- **Shows each patient's appointment count.**
- **Connects the Patients and Appointments tables.**
- **Displays the total appointments for every patient.**

Dispaly the patient details who are not from chennai and mumbai and age>40

```
select*from patients
where city not in('chennai','mumbai')
and  date_of_birth > 40;
```

patient_id	first_name	last_name	city	gender	date_of_birth
1	Ravi	Kumar	Hyderabad	Male	1990-05-12
2	Priya	Sharma	Delhi	Female	1988-08-22
3	Arjun	Reddy	Bengaluru	Male	1995-02-10
6	Neha	Gupta	Pune	Female	1993-03-11
7	Rahul	Das	Kolkata	Male	1991-06-19
8	Anjali	Mehta	Ahmedabad	Female	1996-09-25
9	Kiran	Patel	Surat	Male	1989-12-05
10	Vikram	Singh	Jaipur	Male	1987-04-12
11	Pooja	Nair	Kochi	Female	1994-01-28
12	Suresh	Yadav	Lucknow	Male	1986-02-14
13	Deepak	Joshi	Indore	Male	1990-03-03
14	Meena	Pillai	Trivandrum	Female	1992-04-08
15	Raj	Malhotra	Chandigarh	Male	1988-05-19
16	Simran	Kaur	Amritsar	Female	1995-06-11
17	Nikhil	Jain	Bhopal	Male	1991-07-22
18	Divya	Agarwal	Noida	Female	1993-08-30
19	Manoj	Tiwari	Patna	Male	1984-09-17
20	Kavita	Mishra	Varanasi	Female	1989-10-01
21	Rohit	Chopra	Gurgaon	Male	1992-01-09

- **Displays patient details.**
- **Excludes patients from Chennai and Mumbai.**
- **Shows only patients with age greater than 40.**

show the details of patients who are from Hyderabad.

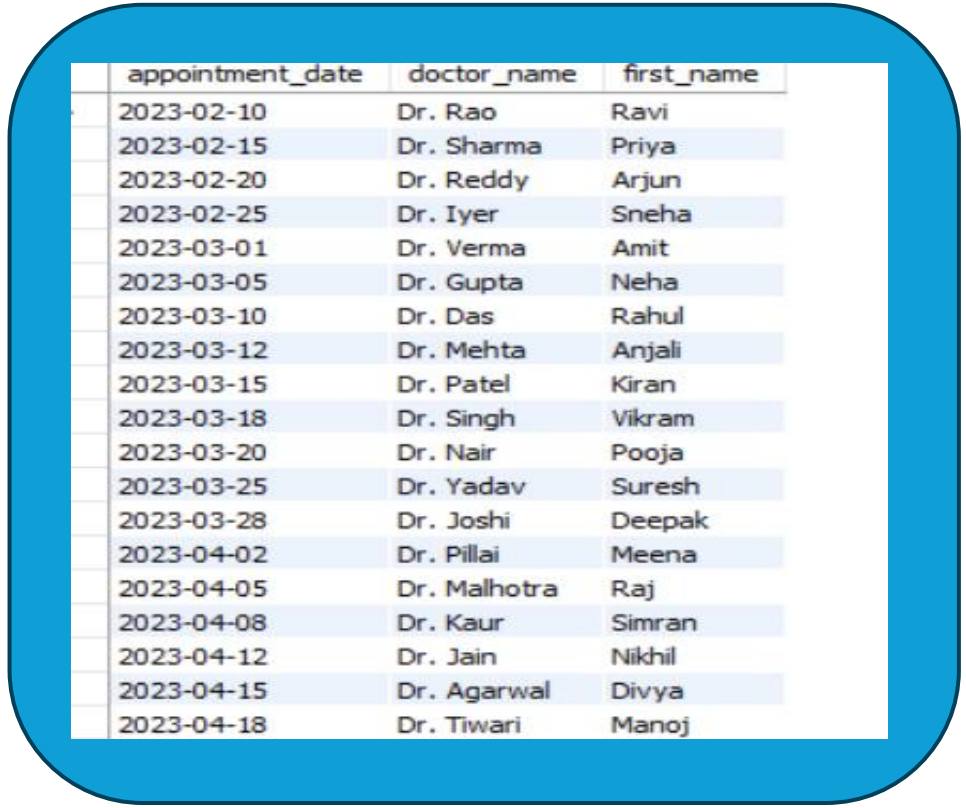
```
select p.*, a.department from patients p inner join appointments a
ON p.patient_id = a.patient_id
where p.city = 'Hyderabad';
```

patient_id	first_name	last_name	city	gender	date_of_birth	department
1	Ravi	Kumar	Hyderabad	Male	1990-05-12	Cardiology
28	Nisha	Rao	Hyderabad	Female	1995-08-12	General
48	Ashok	Rao	Hyderabad	Male	1989-08-23	General

- **The query filters patients based on the city name "Hyderabad".**
- **It retrieves patient and appointment information together.**
- **It returns only the matching records from both tables.**

Display appointment date, doctor name, and patient name.

```
select a.appointment_date,a.doctor_name,p.first_name
from patients p right join appointments a
ON p.patient_id = a.patient_id;
```



appointment_date	doctor_name	first_name
2023-02-10	Dr. Rao	Ravi
2023-02-15	Dr. Sharma	Priya
2023-02-20	Dr. Reddy	Arjun
2023-02-25	Dr. Iyer	Sneha
2023-03-01	Dr. Verma	Amit
2023-03-05	Dr. Gupta	Neha
2023-03-10	Dr. Das	Rahul
2023-03-12	Dr. Mehta	Anjali
2023-03-15	Dr. Patel	Kiran
2023-03-18	Dr. Singh	Vikram
2023-03-20	Dr. Nair	Pooja
2023-03-25	Dr. Yadav	Suresh
2023-03-28	Dr. Joshi	Deepak
2023-04-02	Dr. Pillai	Meena
2023-04-05	Dr. Malhotra	Raj
2023-04-08	Dr. Kaur	Simran
2023-04-12	Dr. Jain	Nikhil
2023-04-15	Dr. Agarwal	Divya
2023-04-18	Dr. Tiwari	Manoj

- **This query displays the appointment date, doctor name, and patient name.**
- **It joins the patients and appointments tables using the patient ID.**
- **The output shows the required details for each appointment.**

Display patients who paid the highest consultation fee.

```sql
WITH MaxFee AS
(
    SELECT MAX(consultation_fee) AS max_fee
    FROM appointments
)
select p.patient_id,p.first_name,p.last_name,a.department,a.consultation_fee
from patients p inner join appointments a
ON p.patient_id = a.patient_id
where a.consultation_fee =
(
    select max_fee from
    MaxFee
    );
```

patient_id	first_name	last_name	department	consultation_fee
44	Kunal	Kapoor	Neurology	1250

- This query finds the highest consultation fee from the appointments table.
- It joins the patients and appointments tables using the patient ID.
- It displays the details of the patient who paid the highest consultation fee.

Assign row numbers to all appointments based on consultation fee (highest first).

```
select appointment_id,doctor_name,department,consultation_fee,
row_number() over(order by consultation_fee DESC) AS row_no
from appointments;
```

appointment_id	doctor_name	department	consultation_fee	row_no
144	Dr. Kapoor	Neurology	1250	1
105	Dr. Verma	Neurology	1200	2
136	Dr. Verma	Neurology	1200	3
125	Dr. Saxena	Neurology	1150	4
113	Dr. Joshi	Neurology	1100	5
145	Dr. Jindal	Gynecology	950	6
114	Dr. Pillai	Gynecology	950	7
133	Dr. Kulkarni	Gynecology	950	8
104	Dr. Iyer	Gynecology	900	9
127	Dr. Shah	Cardiology	900	10
140	Dr. Aggarwal	Cardiology	900	11
120	Dr. Mishra	Gynecology	900	12
146	Dr. Iyer	Cardiology	850	13
115	Dr. Malhotra	Cardiology	850	14
134	Dr. Naik	Cardiology	850	15
121	Dr. Chopra	Cardiology	800	16
101	Dr. Rao	Cardiology	800	17
109	Dr. Patel	Cardiology	800	18
130	Dr. Srivastava	Orthopedics	750	19
117	Dr. Jain	Orthopedics	750	20
123	Dr. Bansal	Orthopedics	700	21

- **This query assigns a row number to each appointment.**
- **The appointments are sorted by consultation fee in descending order.**
- **The highest consultation fee gets row number 1.**

Display patients born between 1990-01-01 and 1995-12-31.

```sql
select *from patients
where date_of_birth BETWEEN '1990-01-01' AND '1995-12-31';
```

patient_id	first_name	last_name	city	gender	date_of_birth
1	Ravi	Kumar	Hyderabad	Male	1990-05-12
3	Arjun	Reddy	Bengaluru	Male	1995-02-10
4	Sneha	Iyer	Chennai	Female	1992-11-01
6	Neha	Gupta	Pune	Female	1993-03-11
7	Rahul	Das	Kolkata	Male	1991-06-19
11	Pooja	Nair	Kochi	Female	1994-01-28
13	Deepak	Joshi	Indore	Male	1990-03-03
14	Meena	Pillai	Trivandrum	Female	1992-04-08
16	Simran	Kaur	Amritsar	Female	1995-06-11
17	Nikhil	Jain	Bhopal	Male	1991-07-22
18	Divya	Agarwal	Noida	Female	1993-08-30
21	Rohit	Chopra	Gurgaon	Male	1992-01-09
24	Swati	Dubey	Nagpur	Female	1991-04-26
25	Harsh	Saxena	Kanpur	Male	1990-05-14
26	Tina	Roy	Kolkata	Female	1993-06-03
28	Nisha	Rao	Hyderabad	Female	1995-08-12

- **This query displays patients born between 1990 and 1995.**
- **It uses the `BETWEEN` operator to filter the date of birth.**
- **Only patients whose birth dates fall within the given range are shown.**

show all appointments with consultation fee greater than 700 and patient details

```
select p.*,a.* from patients p inner join appointments a
ON p.patient_id = a.patient_id
where a.consultation_fee > 700;
```

patient_id	first_name	last_name	city	gender	date_of_birth	appointment_id	patient_id	appointment_date	doctor_name	department	consultation_fee
1	Ravi	Kumar	Hyderabad	Male	1990-05-12	101	1	2023-02-10	Dr. Rao	Cardiology	800
4	Sneha	Iyer	Chennai	Female	1992-11-01	104	4	2023-02-25	Dr. Iyer	Gynecology	900
5	Amit	Verma	Mumbai	Male	1985-07-18	105	5	2023-03-01	Dr. Verma	Neurology	1200
9	Kiran	Patel	Surat	Male	1989-12-05	109	9	2023-03-15	Dr. Patel	Cardiology	800
13	Deepak	Joshi	Indore	Male	1990-03-03	113	13	2023-03-28	Dr. Joshi	Neurology	1100
14	Meena	Pillai	Trivandrum	Female	1992-04-08	114	14	2023-04-02	Dr. Pillai	Gynecology	950
15	Raj	Malhotra	Chandigarh	Male	1988-05-19	115	15	2023-04-05	Dr. Malhotra	Cardiology	850
17	Nikhil	Jain	Bhopal	Male	1991-07-22	117	17	2023-04-12	Dr. Jain	Orthopedics	750
20	Kavita	Mishra	Varanasi	Female	1989-10-01	120	20	2023-04-20	Dr. Mishra	Gynecology	900
21	Rohit	Chopra	Gurgaon	Male	1992-01-09	121	21	2023-04-25	Dr. Chopra	Cardiology	800
25	Harsh	Saxena	Kanpur	Male	1990-05-14	125	25	2023-05-08	Dr. Saxena	Neurology	1150
27	Aditya	Shah	Ahmedabad	Male	1988-07-07	127	27	2023-05-15	Dr. Shah	Cardiology	900
30	Alok	Srivastava	Prayagraj	Male	1991-10-10	130	30	2023-05-25	Dr. Srivastava	Orthopedics	750
33	Preeti	Kulkarni	Pune	Female	1992-03-29	133	33	2023-06-08	Dr. Kulkarni	Gynecology	950
34	Sanjay	Naik	Goa	Male	1985-04-21	134	34	2023-06-12	Dr. Naik	Cardiology	850
36	Akash	Verma	Delhi	Male	1993-06-16	136	36	2023-06-18	Dr. Verma	Neurology	1200
40	Ankit	Aggarwal	Noida	Male	1987-10-05	140	40	2023-07-02	Dr. Aggarwal	Cardiology	900
44	Kunal	Kapoor	Mumbai	Male	1986-04-02	144	44	2023-07-12	Dr. Kapoor	Neurology	1250
45	Payal	Jindal	Panipat	Female	1993-05-25	145	45	2023-07-15	Dr. Jindal	Gynecology	950
46	Ramesh	Iyer	Chennai	Male	1985-06-30	146	46	2023-07-18	Dr. Iyer	Cardiology	850

- **This query joins the patients and appointments tables using the patient ID.**
- **It displays appointments with a consultation fee greater than 700.**
- **It shows both patient details and appointment details in the result.**

Display the first consultation fee in each department.

```
select appointment_id,department,consultation_fee,
first_value(consultation_fee)
over(partition by department order by consultation_fee DESC) AS first_fee
from  appointments;
```

appointment_id	department	consultation_fee	first_fee
127	Cardiology	900	900
140	Cardiology	900	900
146	Cardiology	850	900
115	Cardiology	850	900
134	Cardiology	850	900
101	Cardiology	800	900
121	Cardiology	800	900
109	Cardiology	800	900
122	Dermatology	600	600
131	Dermatology	550	600
149	Dermatology	550	600
111	Dermatology	550	600
102	Dermatology	500	600
137	Dermatology	500	600
116	Dermatology	500	600
143	Dermatology	500	600
135	ENT	650	650
119	ENT	650	650
150	ENT	650	650
107	ENT	600	650
142	ENT	600	650

- **This query displays the first consultation fee in each department.**
- **It uses the `FIRST_VALUE()` window function.**
- **The highest consultation fee in each department is shown as the first fee.**

Display all Delhi patients and doctor names.

```
select p.first_name,a.doctor_name,p.city
from patients p left join  appointments a
ON p.patient_id=a.patient_id
where  p.city='Delhi';
```



first_name	doctor_name	city
Priya	Dr. Sharma	Delhi
Isha	Dr. Kapoor	Delhi
Akash	Dr. Verma	Delhi
Tarun	Dr. Arora	Delhi
Bhavna	Dr. Chawla	Delhi
Sakshi	Dr. Arora	Delhi
Diya	NULL	Delhi
Sana	NULL	Delhi
Zara	NULL	Delhi
Anvi	NULL	Delhi
Arnav	NULL	Delhi
Parth	NULL	Delhi
Pallavi	NULL	Delhi
Kriti	NULL	Delhi
Monika	NULL	Delhi
Pinky	NULL	Delhi
Dinesh	NULL	Delhi
Mukesh	NULL	Delhi
Megha	NULL	Delhi

- **This query displays all patients who live in Delhi.**
- **It joins the patients and appointments tables using the patient ID.**
- **It shows the patient name, doctor name, and city in the result.**

Thank you